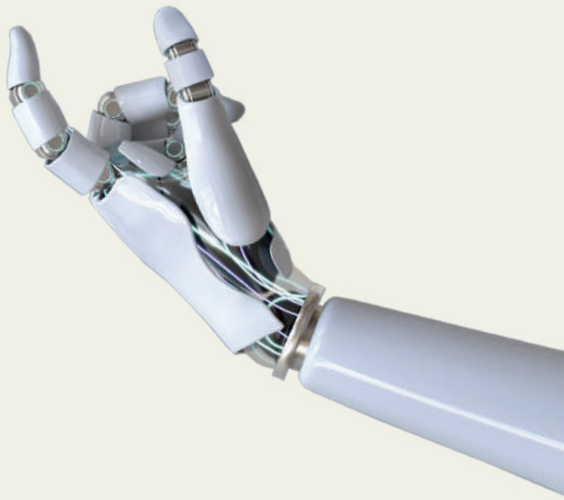
# Vis Decode

AI-driven interpretation
& enhancement of scientific plots



## Kick-off Workshop

December 2024

# Summary

**Vis Decode** project launched its collaborative kickoff workshop on *December 16, 2024*. The event gathered researchers, neuroscientists, data analysts, educators, designers and scientific communicators to co-design an open-source AI tool to support the creation of scientific visualizations. Structured around Design Thinking methodologies, the workshop emphasized participatory problem-solving, with sessions dedicated to defining user needs, prototyping solutions, and outlining a development roadmap ahead of the tool's official release in 2026.

This project aims to create an **AI tool capable of automatically interpreting and providing feedback on scientific plots.** Utilizing state-of-the-art visual language understanding techniques, VisDecode analyzes raster images of plots such as bar charts, line charts, and scatter plots. It extracts key visual attributes like color, shape, positioning, and plot data, all of which significantly impact data perception and understanding. Based on these analyses and well-established best practices from data visualization literature, VisDecode offers actionable suggestions for improving the design of these plots.

The project is led by the Artificial Intelligence Laboratory at Torcuato Di Tella University and supported by the Alfred P. Sloan Foundation (Grant ID: SL-2024-VD009).

## Project Goals

**Build an open-source tool**
*Design and launch a flexible, community-driven platform that provides comprehensive guidance on creating advanced visualizations.*

**Enhance interoperability**
*Ensure integration with existing tools (e.g., Vega-Lite, Python/R libraries) and platforms (e.g., Jupyter, LaTeX).*

**Grow a user community**
*Grow an engaged user community by bringing together a diverse, interdisciplinary network of researchers around the tool.*

**Provide educational resources**
*Provide comprehensive documentation, tutorials, and case studies to support effective tool usage.*

**Support visualization creation**
*Develop an AI model to support the creation of scientific visualizations, emphasizing alignment with disciplinary standards.*

# Background

## Why Visualization Matters

For over three centuries, charts and graphs have been indispensable for translating quantitative data into actionable insights. From Florence Nightingale's 1853 "rose diagram"—which revolutionized public health policy by exposing preventable wartime deaths—to modern climate science, effective visualizations can uncover patterns, challenge assumptions, and drive discovery. Anscombe's Quartet exemplifies this: four datasets with identical statistical properties reveal radically different relationships only when visualized, underscoring the limitations of summary metrics and the necessity of graphical analysis [Anscombe 1973].

## The Challenge of Designing Effective Visualizations

Creating visuals demands expertise in both data interpretation and design principles. Even minor choices—color palettes, axis scaling, or chart type—profoundly influence how information is perceived. While tools like Tableau or Vega-Lite simplify encoding data into visual attributes (e.g., position, color), they require users to navigate complex trade-offs. For example, Figure x contrasts a cluttered NIH budget pie chart (misleading angular comparisons, chaotic labels) with a redesigned bar chart (sorted, perceptually optimized), illustrating how poor design obscures insights [Savva 2011].

## AI as a Collaborative Partner in Visualization Design

Recent advances in AI offer transformative potential. Systems like VisDecode aim to automate design critiques and suggest improvements while preserving user agency. Unlike proprietary tools, VisDecode prioritizes interoperability with open standards like Vega-Lite, enabling researchers to refine visuals within their existing workflows (LaTeX, Python) and allow access to expert-level visualization principles feedback.

# Workshop Design

The main goal of the kickoff workshop was to align all stakeholders on the vision, scope, and roadmap of VisDecode by establishing a shared understanding of the project's mission and expected outcomes.

During the event, participants also clarified their roles and responsibilities, identified key challenges and research questions, and outlined the methodology, technological framework, timeline, and key milestones.
The workshop was structured around three main themes derived from pre-event surveys and the project's core objectives:

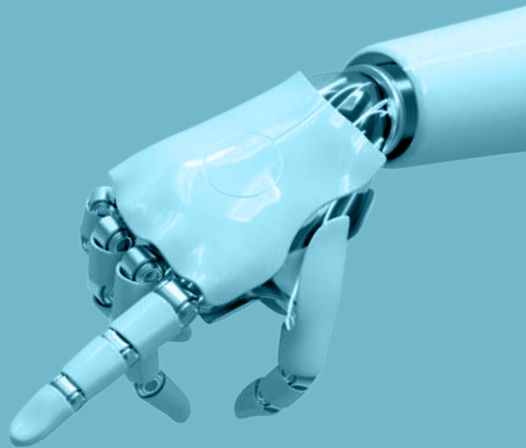
- **User needs**
  *Understanding and capturing the real-world requirements, constraints, and pain points faced by users when creating visualizations.*

- **Investigating interface designs**
  *Prototyping user flows to ensure a seamless and engaging experience tailored to diverse usage contexts.*

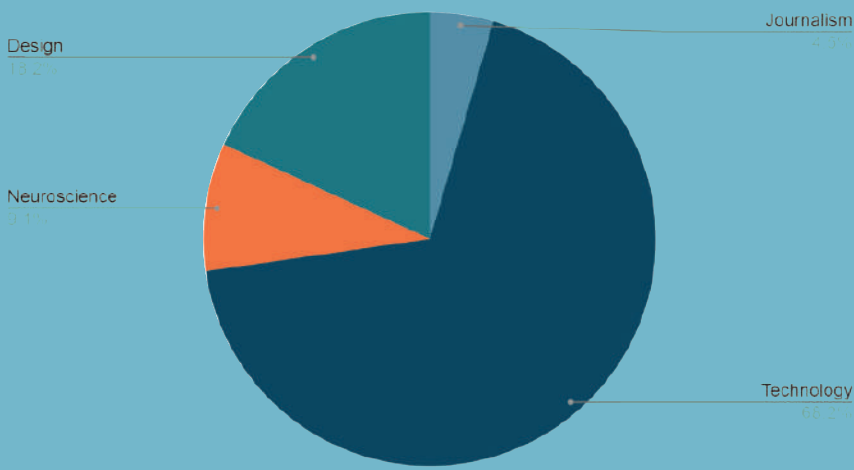
- **Technical solutions**
  *Exploring the AI technologies best suited to power the software, while identifying the relevant resources needed for implementation.*

# Participants

The workshop brought together 22 participants from diverse professional backgrounds, ensuring a multidisciplinary approach to the challenges of data visualization. The majority came from the technology sector, while the remaining participants represented design , neuroscience, and journalism . Additionally, five of the attendees were professors and researchers, contributing academic perspectives to the discussion.



| **Background** | **Amount** | **Representation** |
|---|---|---|
| Journalism | 1 | 4,5% |
| Technology | 15 | 68,2% |
| Neuroscience | 2 | 9,1% |
| Design | 4 | 18,2% |




This diverse composition fostered rich exchanges of ideas, blending technical expertise with insights from design, neuroscience, and communication.

# Workshop Methodology & Execution

The workshop was structured into two key moments across four interdisciplinary working groups. Each group was intentionally balanced to ensure a diverse blend of professionals from different backgrounds, encouraging a variety of perspectives on the subject.

## Agenda

- 10:00 - 10:30 | **Presentation of the project and workshop objectives**
- 10:30 - 11:00 | **First group work session.**
- 11:00 - 11:20 | **First round of presentations - 5 minutes per group**
- 11:20 - 11:50 | **Second group work session**
- 11:50 - 12:10 | **Second round of presentations - 5 minutes per group**
- 12:10 - 12:45 | **Lunch break**
- 12:45 - 14:00 | **Closing session and roadmap definition**

## Introduction

The session began with an overview of the importance of data visualization, highlighting the need to improve accessibility and usability. Producing effective data visualizations is a complex process that involves multiple decisions: selecting and filtering the right data, applying design principles, ensuring accessibility best practices, mastering technical tools and programming languages, and dedicating significant time to achieving high-quality results. Participants were introduced to key challenges in the field and the role of AI-driven tools in supporting more effective visual communication. The facilitators also outlined the importance of collaboration and iterative problem-solving.

## Group Activity

Participants worked in interdisciplinary groups of 4-5 people, using sticky notes to document their ideas and facilitate discussion. The exercise was divided into two moments: 30-minute work sessions, each followed by a round of presentations -5 minutes per group-, allowing for discussion and feedback before moving on to the next phase.

1. **Defining users and problems:**
Before starting, participants were provided with the following guiding questions to structure their analysis:

.*Who are the typical users of data visualization tools?*
.*What are their levels of experience and fields of work?*
.*What are the most common frustrations or obstacles they face?*

During this phase, teams identified primary user profiles and detailed the challenges they face when creating data visualizations. They described user demographics (age, experience level, field of work) and listed common issues such as unfamiliarity with design principles, technical barriers, or the lack of accessible tools.

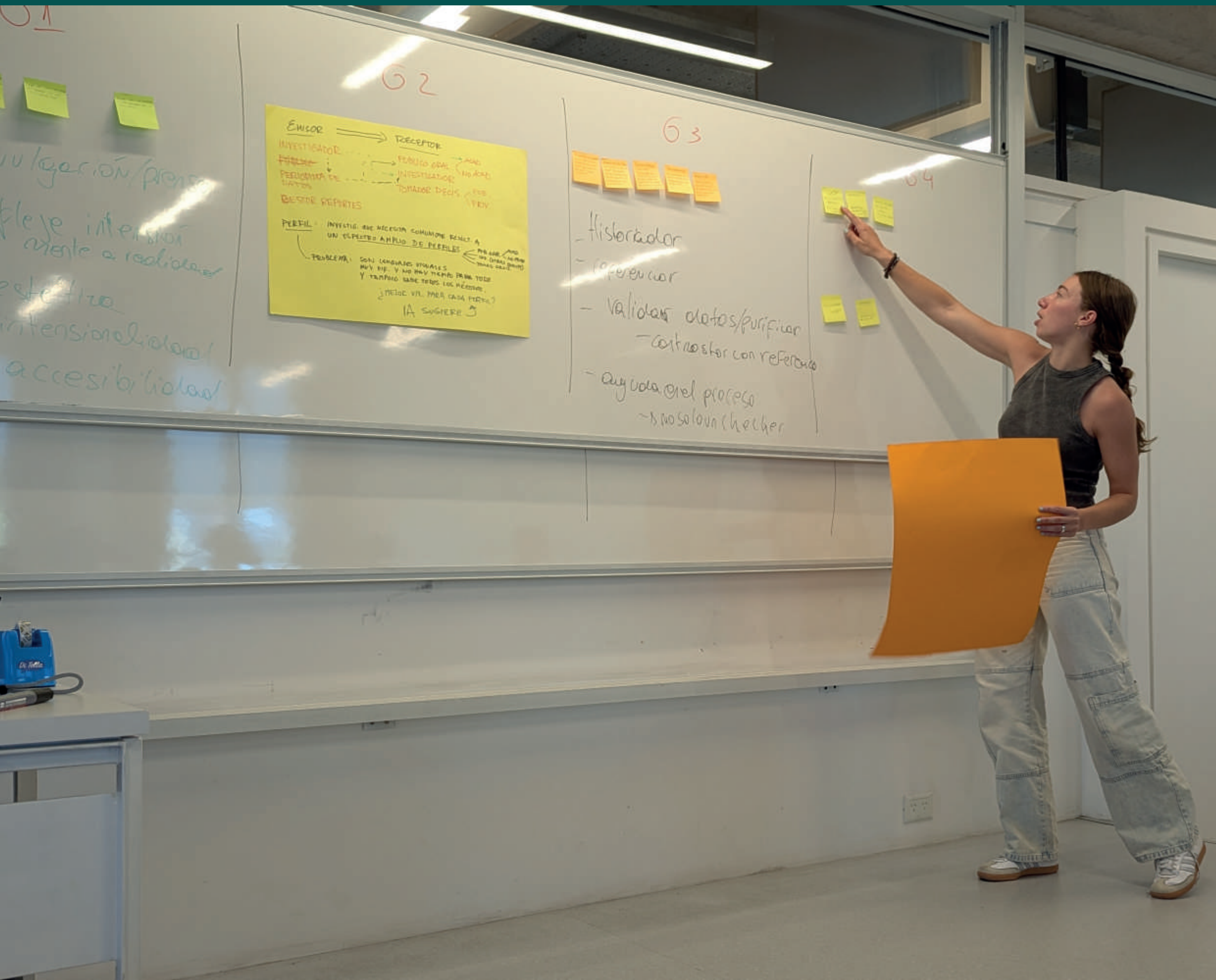
## 2. Designing solutions and interfaces:

In the second phase, participants used the following questions as a framework to propose and develop potential solutions:

*.What kind of tool or system could address the identified problems?*
*.What technical challenges would need to be solved?*
*.What visualization issues should the tool be capable of identifying?*
*.How would this solution fit into a typical user's workflow?*

Based on the identified challenges, teams conceptualized possible solutions, including interface designs and AI-driven feedback mechanisms to assist users. Some groups also sketched user interaction flows to illustrate their ideas.

After each round of presentations, a collective discussion was encouraged, allowing participants to exchange perspectives, refine their ideas, and build upon one another's insights.



# Outcomes

Collaborative activities revealed two distinct user profiles shaping the tool's design priorities. **Researchers** emphasized the need for solutions that simplify complex visualization tasks without compromising academic rigor, calling for iterative feedback mechanisms, accessibility compliance (e.g., color-blind-friendly palettes, alt-text generation), and integration with established workflows like LaTeX and Python.

In contrast, **communicators and designers** (e.g., journalists, educators) prioritized intuitive, code-free interfaces to enable rapid experimentation, emphasizing storytelling-driven design principles and tools that harmonize precise data representation with visually compelling narratives. These insights underscored the project's dual focus on technical precision for domain experts and user-centric agility for broader audiences.

## Challenges Identified

**Technical adequacy vs. user experience gap**
*Existing tools excel at rendering data but neglect the human side of visualization. Users report frustration with tools that solve technical plotting tasks but fail to guide design decisions or simplify workflows.*

**Balancing simplicity and customization**
*Participants demanded tools that offer smart defaults like - perceptually sound color palettes, chart types- while retaining flexibility for advanced tweaks. Current tools force a false choice between ease-of-use and control.*

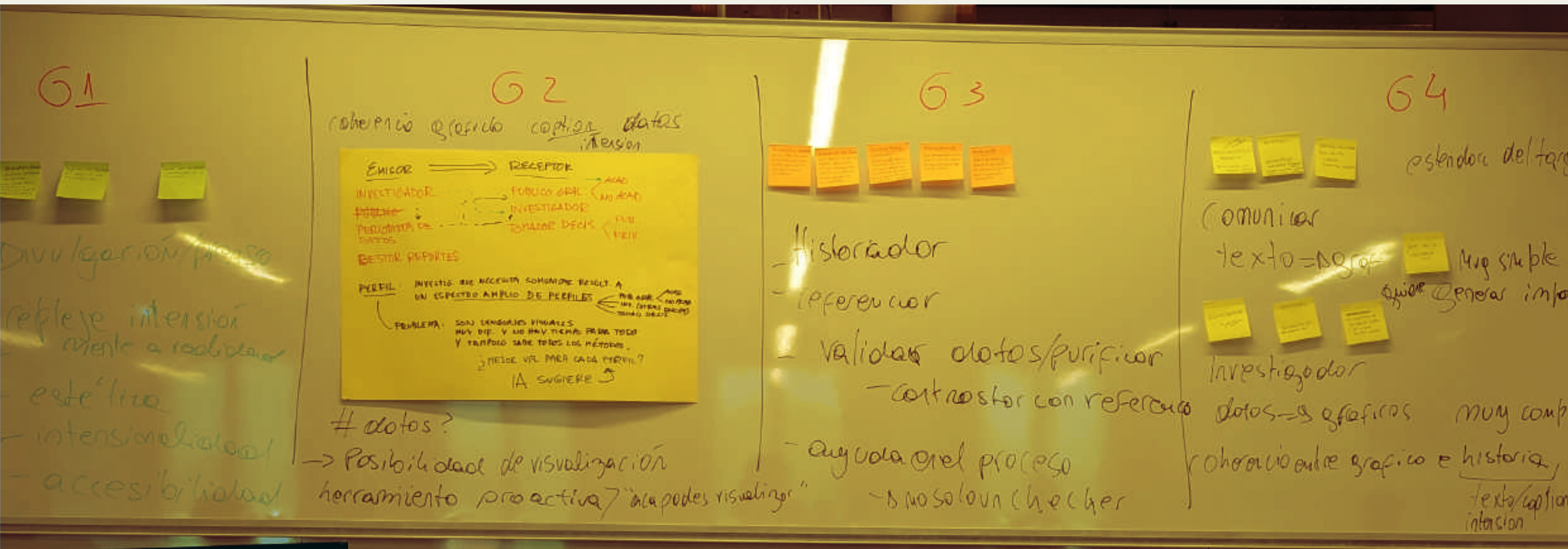
**Democratizing design expertise**
*Non-experts struggle to translate ideas into visuals due to a lack of design literacy. Tools assume users know what to visualize and how to encode data effectively.*

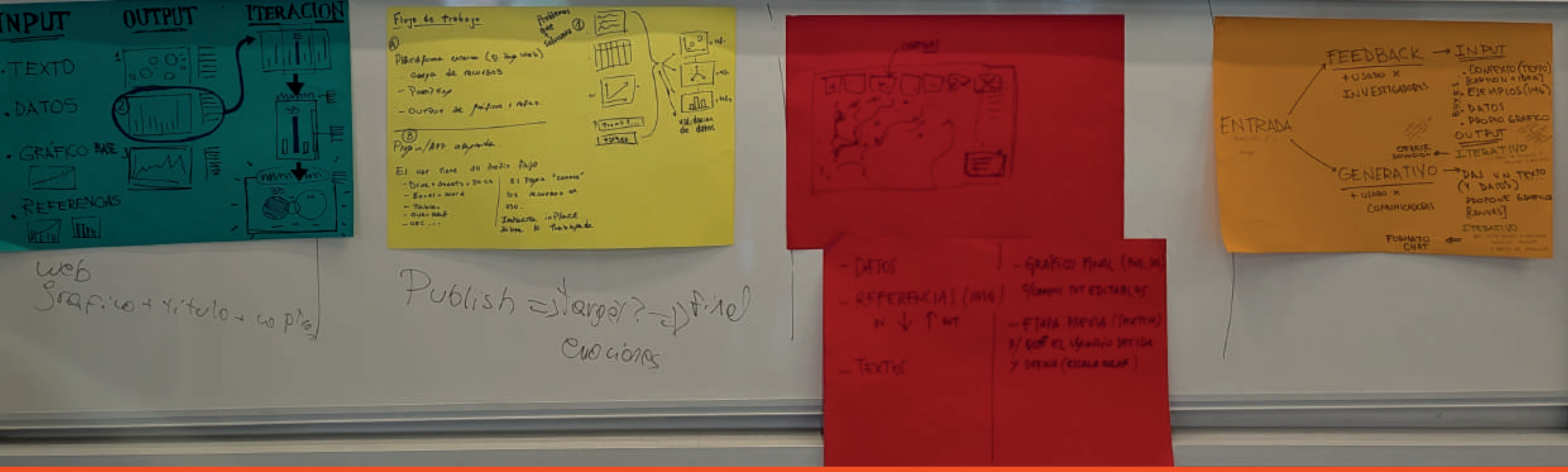
**Making interactivity accessible**
*Interactive features are often abandoned due to steep learning curves. Workshop participants suggested interactivity should enhance clarity, not complexity.*

**Sustainable open ecosystems**
*Many current tools for creating charts and graphs are either "locked" or "disconnected". This makes it hard for people to share ideas, reuse designs, or improve tools as a team.*

The following chart summarizes how each group approached each proposed exercise:

| Group | Defining Users & Problems | Designing Solutions & Interfaces |
|---|---|---|
| 1 | **Users:** Science communicators, journalists and media professionals.<br><br>**Challenges:** Developing an easy-to-use tool that prioritizes aesthetics and effectively conveys the intended message. | **Solution:** The interface should allow users to input text, existing graphics, datasets, and style references. The AI should generate visualization options with explanatory text justifying design choices. Users should have the ability to select an option and request iterations, with or without new instructions. |
| 2 | **Users:** Researchers and data journalists aiming to communicate findings to peers, general public, and policymakers.<br><br>**Challenges:** Adapting communication styles for different audiences; ensuring AI can recognize audience-specific needs and adjust outputs accordingly. | **Solution:** The tool could be a local plugin integrated into existing software or an external web application. The interface should accept text-based prompts, existing graphics, datasets, and style references. Based on the inputs, the AI should generate multiple visualization options and provide references to the sources that informed its design decisions. |
| 3 | **Users:** Subject-matter experts with limited visualization expertise.<br><br>**Challenges:** AI should not only validate and correct but also propose visualization approaches, as users may lack knowledge of best practices. Feedback should be evidence-based. | **Solution:** The interface should accept text-based prompts, datasets, and style references. The unique feature of this approach is an iterative sketching system, allowing users to quickly validate AI-generated suggestions before arriving at a final visualization. The interface should display decision trees that show how different choices evolve based on user input. |
| 4 | **Users:** Researchers and journalists, with distinct visualization needs.<br><br>**Challenges:** Researchers must make complex topics accessible while ensuring their message remains clear and compelling. They should also appeal to emotions and convey simple, direct ideas. | **Solution:** The interface should start by distinguishing between user types—researcher or journalist. For researchers, the input should include context, style references, datasets, and existing graphics, with an output that visually represents the decision-making process.<br>For journalists, the input should be an article or academic text, with the AI generating visualization options that can be refined through a chat-like interaction. |






## Strategic Direction

To address these challenges, VisDecode should prioritize:

**AI-Guided design critiques**
*Automatically flag issues and suggest improvements while positioning AI as a collaborator, not a replacement—ensuring users retain control while benefiting from expert-guided support. AI recommendations should be explainable, grounding design choices in visualization theory.*

**Modular workflows**
*Enable users to start with templates and smart defaults, progressively customizing through intuitive interfaces. Ensure flexibility to accommodate different user needs, whether for structured analysis or storytelling-driven visualization.*

**Open standards, UI, & community co-design**
*Build on Vega-Lite for interoperability and actively engage users in tool evolution via open forums, ensuring adaptability and responsiveness to real-world needs. Prioritize compatibility with both standalone use (e.g., web apps) and embedded workflows (e.g., plugins), allowing seamless adoption across various research and communication environments.*